

AYUSH GIRI

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Education

SRM University AP, Amaravati, India
B.Tech in Computer Science & Engineering

2023 – 2027
CGPA: 8.35/10

Technical Skills

Core: Python, Java, C++, JavaScript, TypeScript, SQL

Full-Stack: React, Next.js, FastAPI, Node.js, Express.js, RESTful API Design, JWT Authentication

Databases: PostgreSQL, MongoDB, SQLAlchemy (ORM)

Infra & Tooling: Docker, Redis, Celery, Socket.IO, GitHub Actions (CI), pytest

AI/LLM: Google Gemini API, Prompt Engineering, Schema-Validated LLM Pipelines

Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks

Research Experience

Research Intern

Aug 2025

SRM University AP

- Engineered distributed computation workflows and feature extraction frameworks, improving the efficiency of machine learning experimentation.
- 50+ experiments automated and tracked, achieved by building distributed computation workflows with reproducibility-first experiment logging.
- 8% model accuracy improvement, achieved by re-engineering feature extraction pipelines for CIFAR-10 image classification experiments.

Projects

Nexora : AI-Powered Networking CRM Platform

[Link]

React 19 • Vite • Tailwind CSS • FastAPI • PostgreSQL • SQLAlchemy • Celery • Redis • Docker

- Zero unauthorized access incidents across **dual JWT/Supabase** auth modes, achieved by architecting role-based access control with rate limiting across **310+ REST** endpoints.
- 100% environment parity across **dev/prod**, achieved by containerizing the full stack (React, FastAPI, PostgreSQL, Redis) with Docker Compose.
- Automated regression coverage on all critical auth/API paths, achieved by building a CI/CD-ready pytest + Playwright pipeline.

Aether : Multi-Agent React UI Generation System

[Link]

Next.js 14 • TypeScript • React • Zod • Google Gemini API

- Enforced output reliability on every API route and a custom prompt-injection guard to sanitize LLM-facing input.
- 100%** schema-validated outputs, eliminating malformed LLM responses, by enforcing **Zod validation** and a custom prompt-injection guard across every API route.
- Zero downtime during live API outages, by building a mock-mode fallback enabling fully offline development and demos.

CodeForge : Real-Time Collaborative Code Editor

[Link]

React • Node.js • Express.js • MongoDB • Socket.IO • JWT • Judge0 API

- Built a MERN-stack collaborative code editor supporting real-time multi-user editing and room-based sessions.
- Less than **5% sandbox escapes** in testing, by integrating secure, isolated code execution via a protected `/api/execute` endpoint on **Judge0 CE**.
- Real-time sync across concurrent multi-user sessions, by implementing Socket.IO with room-based, JWT-authenticated collaboration.

Certifications

MongoDB Certified Associate Developer

MongoDB Inc

Validated skills in schema design, aggregation pipelines, indexing strategies, and CRUD operations on document-oriented databases.

Java SE 17 Developer

Oracle

Validated proficiency in core Java, OOP design, exception handling, generics, and concurrency.